

COMPUTING MACHINERY AND INTELLIGENCE

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1. The Imitation Game

I propose to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous.

The new form of the problem can be described in terms of a game which we call the 'imitation game'. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman.

We now ask the question, 'What will happen when a machine takes the part of A in this game?' Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, 'Can machines think?'

2. Critique of the New Problem

As well as asking, 'What is the answer to this new form of the question', one may ask, 'Is this new question a worthy one to investigate?' This latter question we investigate without further ado, by considering the advantages of the imitation game.

The imitation game has the advantage of drawing a fairly sharp line between the physical and the intellectual capacities of a human being. No engineer or chemist claims to be able to produce a material which is indistinguishable from the human skin. It is possible that at some time this might be done, but even if this invention were available, it would be of little interest to try to make a 'thinking machine' more human by dressing it up in such artificial flesh.

3. The Digital Computers

The idea behind digital computers may be explained by saying that these machines are intended to carry out any operations which could be done by a human computer. The human computer is supposed to be following fixed rules; he has no authority to deviate from them in any detail.

We may suppose that these rules are supplied in a book, which is altered whenever he is put on to a new job. He has also an unlimited supply of paper on which he does his calculations.

A digital computer can usually be regarded as consisting of three parts:

- (i) Store of information.
- (ii) Executive unit which carries out operations.
- (iii) Control which ensures instructions are obeyed correctly.

4. Universality of Digital Computers

Digital computers can be constructed, and indeed have been constructed, according to these principles. They can in fact mimic the actions of any other discrete state machine.

This special property of digital computers, that they can simulate any discrete state machine, is described by saying that they are universal machines. The existence of machines with this property has the important consequence that, considerations of speed apart, it is unnecessary to design various new machines to do various computing processes. They can all be done with one digital computer, suitably programmed for each case.